A Mini Project Report Of

PCC-201-COM : Data Structures

Submitted by

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CERTIFICATE

This is to certify that the "**Mini Project Report**" submitted by **Prathamesh Ganesh Bhute (48),** on **"Implementation of Palindrome checker [Stacks and Queues]"** is work done by her and submitted during 2026–2027 academic year, in partial fulfillment of the requirements for the award of the degree of **Bachelor of engineering** in computer Engineering, at **Bharati Vidyapeeth's College of Engineering, Lavale, Pune**.

Prof. Tejal H. Patil
(Guide)

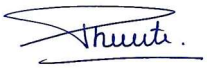
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Head of Computer Engg. Department

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Prathamesh Ganesh Bhute

ABSTRACT

The mini project "**Implementation of Palindrome checker Using [Stacks and Queues]**" aims to demonstrate the practical application of data structures in solving real-world problems. A palindrome is a word, phrase, or number that reads the same forward and backward. In this project, the stack data structure is used to reverse the input sequence, while the queue is used to preserve the original order. By comparing elements from both structures, the program determines whether the given input is a palindrome. The implementation highlights core data structure concepts such as LIFO (Last-In-First-Out) in stacks and FIFO (First-In-First-Out) in queues. This project not only reinforces the understanding of stack and queue operations in Python but also enhances problem-solving and algorithmic thinking skills. This project serves as an excellent example of applying data structure concepts to real-world problem-solving using Python programming. It enhances logical thinking, programming skills, and the ability to apply theoretical knowledge of stacks and queues to build efficient and functional software solutions.

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1. INTRODUCTION

1.1 Project Title :-

Palindrome checker Using [Stacks and Queues]

1.2 Introduction :-

A palindrome is a word, number, or string that reads the same forward and backward. For example, MADAM, LEVEL, RADAR, and 121 are palindromes. Words such as HELLO and COMPUTER are not palindromes because they read differently when reversed.

The Palindrome Checker using Stacks and Queues is a Data Structures mini-project developed to check whether a given word or string is a palindrome. This project demonstrates the practical use of two important data structures: Stack and Queue.

A stack is a linear data structure that follows the Last In, First Out (LIFO) principle. This means that the last character inserted into the stack is the first character removed. A queue is a linear data structure that follows the First In, First Out (FIFO) principle. This means that the first character inserted into the queue is the first character removed.

In this project, the user enters a word or string. The characters are inserted into both the stack and the queue. The stack removes characters in reverse order, while the queue removes them in their original order. The removed characters are then compared one by one.

If all the compared characters are the same, the given word or string is a palindrome. If any characters are different, it is not a palindrome. For example, when the user enters `MADAM`, the characters match in both directions, so the system displays that the word is a palindrome. When the user enters `HELLO`, the characters do not match in both directions, so the system displays that it is not a palindrome.

This project can be implemented using Python and basic programming concepts. It helps students understand stack operations such as push and pop, queue operations such as enqueue and dequeue, and the importance of data structures in solving programming problems.

1.3 Objectives of the Project :-

The main objective of this project is to implement and demonstrate the practical use of Stack and Queue data structures through a simple application that checks whether a given word, string, or number is a palindrome. The project aims to improve the conceptual and practical understanding of data structures using Python programming.

The specific objectives are as follows:

1. To understand and implement basic data structures, particularly Stack (LIFO) and Queue (FIFO), using Python.
2. To design an algorithm that uses stack and queue operations to determine whether an input string or number is a palindrome.
3. To demonstrate the differences between stack and queue operations and understand how they work together to solve programming problems.
4. To improve problem-solving and logical thinking skills by applying data structure concepts to a practical application.
5. To implement and understand stack operations such as push and pop, and queue operations such as enqueue and dequeue.

1.4 Scope :

The scope of the Palindrome Checker using Stacks and Queues project is to develop a simple Python program that checks whether a given word, string, or number is a palindrome using stack and queue data structures.

The project focuses on the following areas:

1. **Palindrome Checking**: Checks whether the given input reads the same forward and backward.

2. **Stack Implementation**: Uses stack operations such as push and pop to process characters in reverse order.

3. **Learning Data Structures**: Helps students understand the practical applications of stacks, queues, and their operations.

4. **Character Comparison**: Compares characters obtained from the stack and queue to determine whether the input is a palindrome.

5. **Future Improvements:** The system can be extended to ignore spaces, punctuation, and differences between uppercase and lowercase letters while checking palindromes.

2. LITERATURE SURVEY

A literature survey is a study of existing books, research papers, and other references related to a project. For the Implementation of Palindrome Checker using Stacks and Queues, the literature survey focuses on the concepts of stack and queue data structures, their operations, and their use in checking palindromes.

The study of these concepts helps in understanding how characters can be processed in forward and reverse order to determine whether a given string is a palindrome. The following references provide the theoretical foundation for this project.

Year	Paper Name / Reference	Author	Conclusion
2023	Data Structures and Algorithms Using Python	Subrata Saha	Explains data structures and algorithms using Python, with examples that support understanding stack and queue operations.
2023	Data Structures & Algorithms in Python	John Canning, Alan Broder, and Robert Lafore	Covers data structures and algorithms that help explain the organization and processing of data.
2024	Data Structures and Algorithms with Python: With an Introduction to Multiprocessing	Kent D. Lee and Steve Hubbard	Presents fundamental and advanced data structures, algorithms, and programming examples.
2024	Python 3 Documentation: Data Structures	Python Software Foundation	Explains how Python lists can implement stacks and how queues can be implemented using suitable data structures.
2024	Python 3 Documentation: collections — Container Datatypes	Python Software Foundation	Describes `collections.deque`, which supports efficient insertion and removal at both ends and is useful for queue implementation.

3. UML DIAGRAMS

3.1 UML CLASS DIAGRAM :-

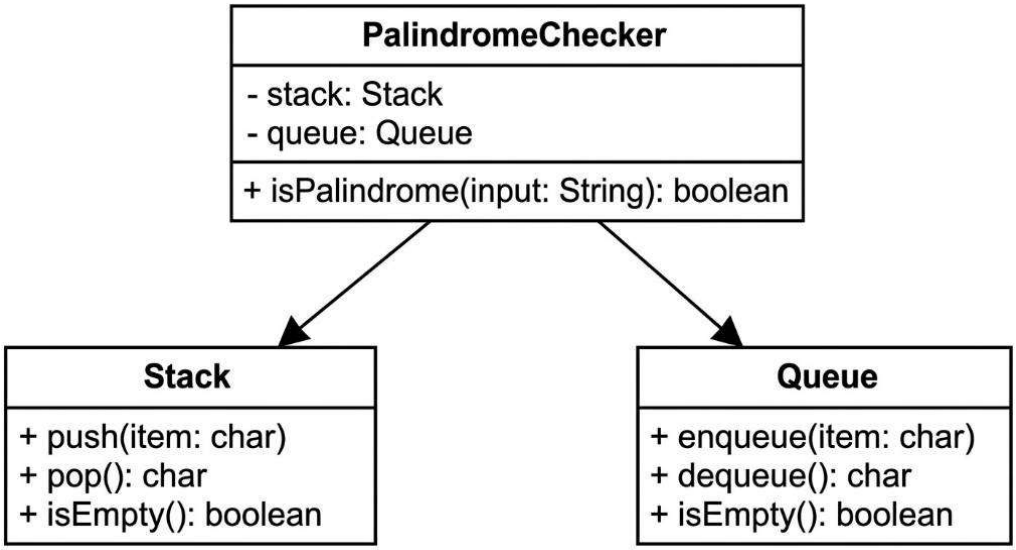


Fig. 3.1: UML Class Diagram of Palindrome Checker Using Stack and Queue

3.2 USE CASE DIAGRAM :-

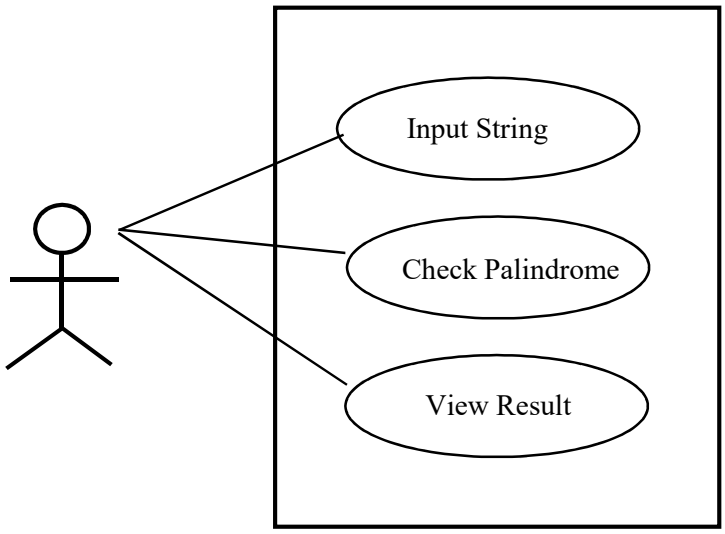


Fig. 3.2 Use Case Diagram of Palindrome Checker Using Stack and Queue

4. SYSTEM REQUIREMENT

4.1 HARDWARE REQUIREMENTS :-

1. **Processor Type**: Intel Core i3 or equivalent processor.
2. **Recommended Processor**: Intel Core i3 or higher.
3. **RAM**: Minimum 2 GB (4 GB recommended).
4. **Processor Speed**: 2.30 GHz or higher recommended.
5. **Hard Disk**: At least 100 MB of free space for Python, source code, and project files.
6. **Input Devices**: Keyboard and mouse.
7. **Output Device**: Monitor or laptop display.

4.2 SOFTWARE REQUIREMENTS :-

1. **Programming Language**: Python 3.8 or above.
2. **IDE / Code Editor**: Jupyter Notebook.
3. **Libraries Used**: Python standard library only. No external libraries are required.
4. **Operating System**: Windows 10/11, Linux, or macOS.

5. SYSTEM DESIGN AND ALGORITH

5.1 Existing System :-

In the existing system, palindrome checking is usually performed by reversing a word, string, or number and comparing it with the original input. If both are the same, the input is a palindrome. Otherwise, it is not a palindrome.

This method is simple and easy to understand. However, it does not necessarily demonstrate the use of data structures such as Stack and Queue.

Limitations of the Existing System:

- It may use string-reversing methods instead of data structures.
- It does not clearly demonstrate Stack and Queue operations.
- It provides limited practical understanding of LIFO and FIFO concepts.
- It may not be suitable for demonstrating Stack and Queue operations in a Data Structures mini-project.

5.2 Proposed System :-

The proposed system is a Palindrome Checker using Stack and Queue developed in Python. It checks whether a given word or number reads the same forward and backward. In this system, each character is stored in both a Stack and a Queue. The Stack follows LIFO (Last In, First Out), while the Queue follows FIFO (First In, First Out). The characters removed from both data structures are compared to check whether the input is a palindrome.

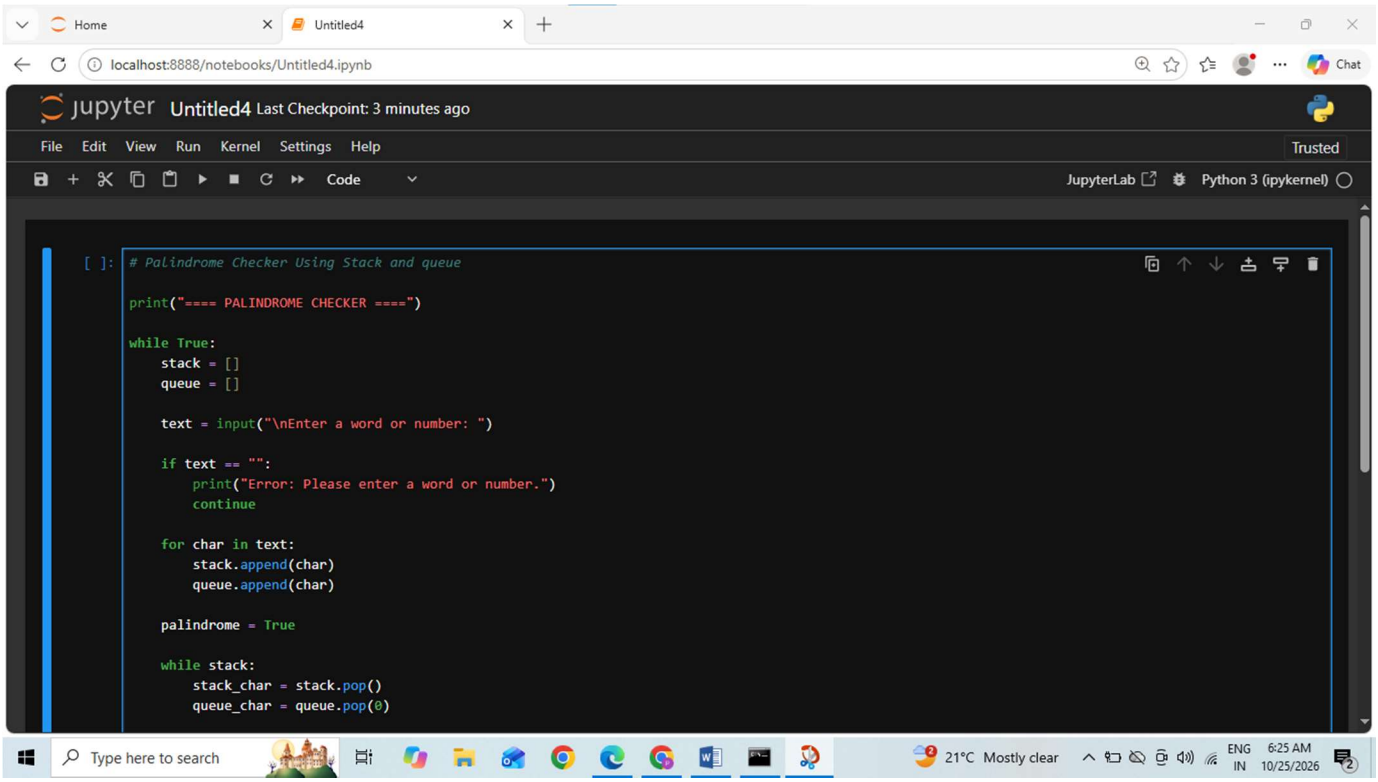
5.3 Algorithm :-

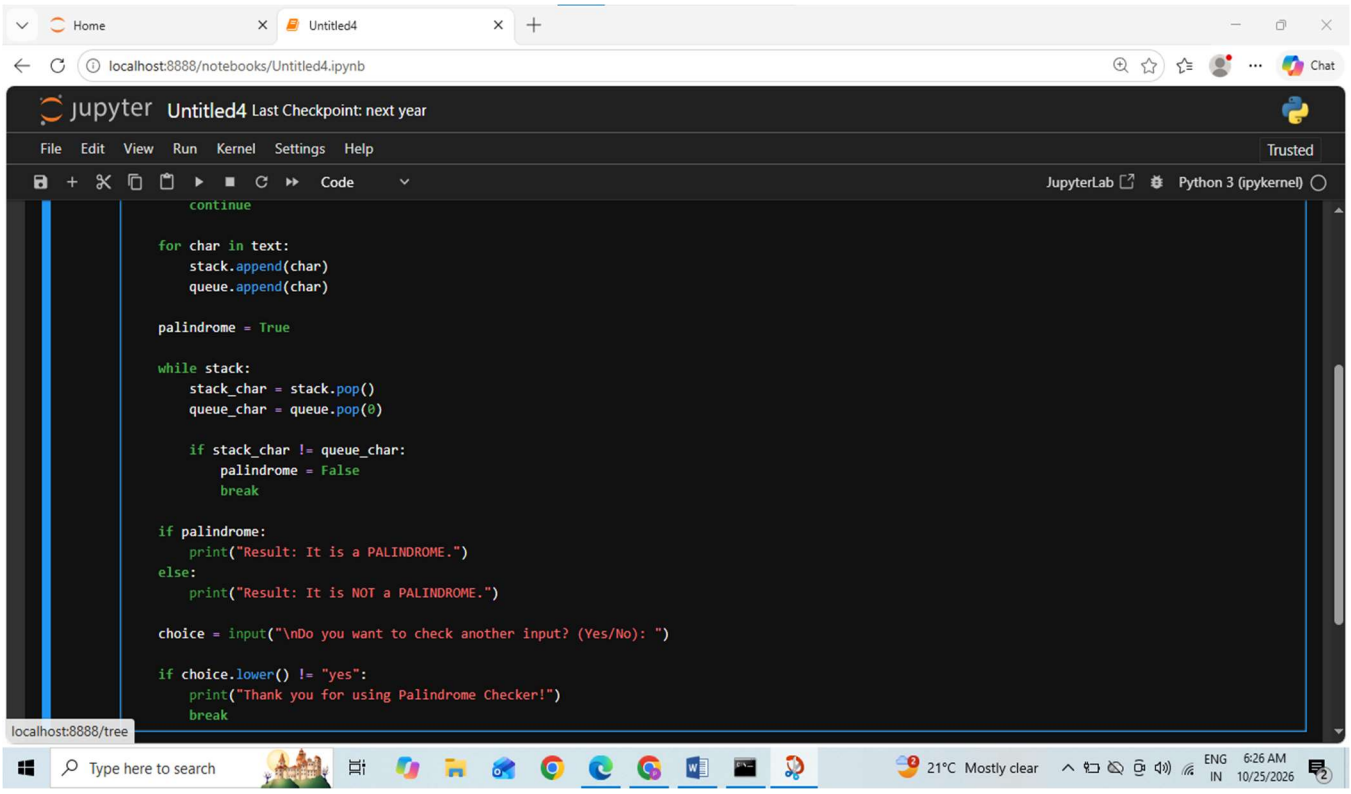
Algorithm: Palindrome Checker Using Stack and Queue

1. Start the program.
2. Display the heading “PALINDROME CHECKER”.
3. Create an empty Stack and Queue.
4. Accept a word or number from the user.
5. If the input is empty, display an error message and ask for input again.
6. Insert each character of the input into both the Stack and Queue.
7. Set `palindrome = True`.
8. Remove one character from the Stack using `pop()` and one from the Queue using `pop(0)`.
9. Compare the two characters.
    - If they are different, set `palindrome = False` and stop comparing.
    - Otherwise, continue until the Stack becomes empty.

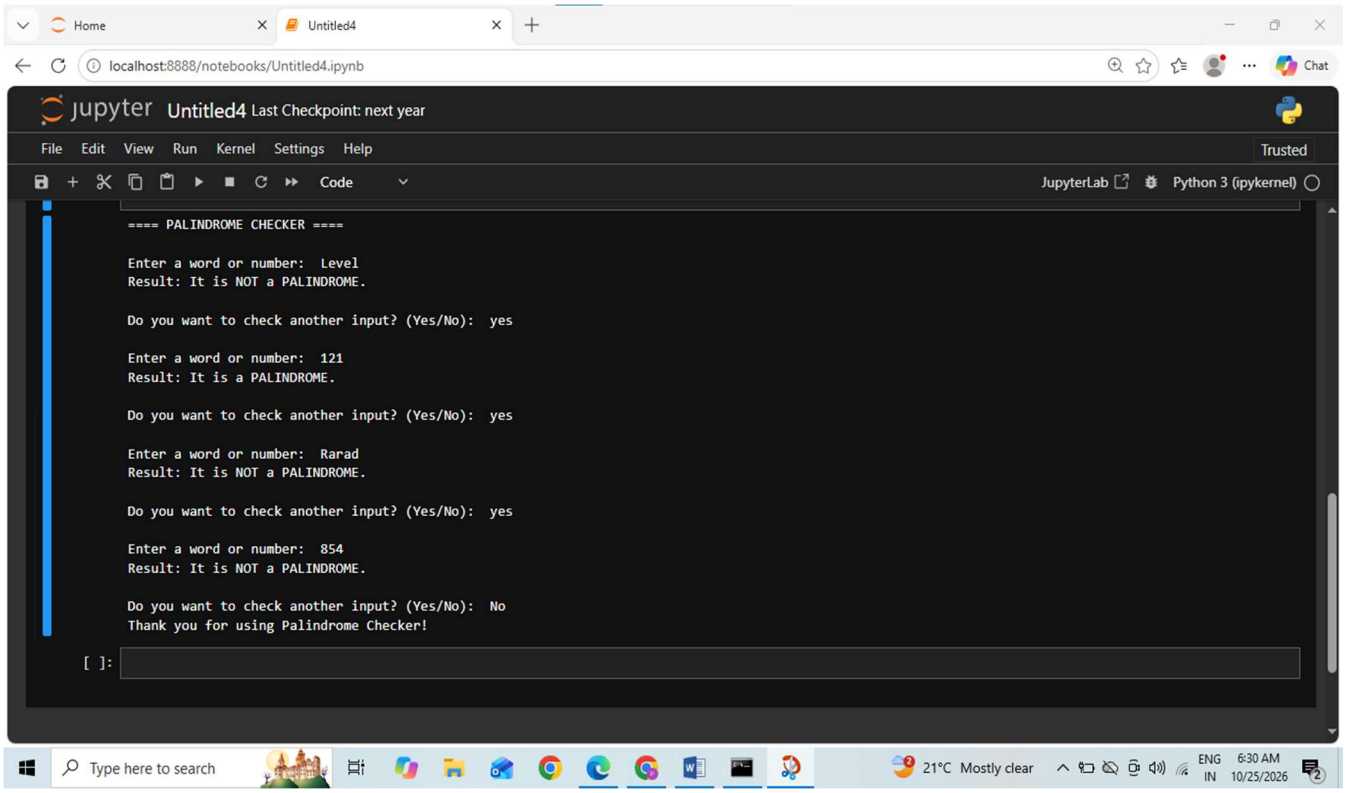
10. Display the result:
    - If `palindrome = True`, display “It is a PALINDROME.”
    - Otherwise, display “It is NOT a PALINDROME.
11. Ask the user whether they want to check another input.
12. If the user enters Yes, repeat the process from Step 3.
13. Otherwise, display the thank-you message.
14. Stop the program.

6. SYSTEM CODE





RESULT


Home
Untitled4
localhost:8888/notebooks/Untitled4.ipynb
Chat
jupyter Untitled4 Last Checkpoint: next year
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
==== PALINDROME CHECKER ====
Enter a word or number: Level
Result: It is NOT a PALINDROME.
Do you want to check another input? (Yes/No): yes
Enter a word or number: 121
Result: It is a PALINDROME.
Do you want to check another input? (Yes/No): yes
Enter a word or number: Rarad
Result: It is NOT a PALINDROME.
Do you want to check another input? (Yes/No): yes
Enter a word or number: 854
Result: It is NOT a PALINDROME.
Do you want to check another input? (Yes/No): No
Thank you for using Palindrome Checker!
[ ]:
Type here to search
21°C Mostly clear
ENG IN
6:30 AM 10/25/2026

7. APPLICATIONS

The Palindrome Checker using Stack and Queue has the following applications:

1. **Word Checking:** Used to check whether a word, such as `MADAM`, reads the same forward and backward.
2. **Number Checking**: Used to check whether a number, such as `12321`, is a palindrome.
3. **String Processing**: Used to check whether a given string is a palindrome.
4. **Learning Data Structures**: Helps students understand Stack (LIFO) and Queue (FIFO) operations.
5. **Programming Practice**: Helps beginners learn Python loops, lists, conditions, and user input.
6. **Text Processing**: The palindrome-checking concept can be used in applications that analyze character sequences.

8. FUTURE SCOPE

The Palindrome Checker using Stack and Queue can be improved in the future by adding the following features:

1. **Graphical User Interface (GUI)**: Develop a simple interface to make the program easier to use.
2. **Case-Insensitive Checking**: Treat uppercase and lowercase letters as the same.
3. **Sentence Checking**: Allow users to check complete sentences, ignoring spaces and punctuation.
4. **Improved Performance**: Optimize the program to check longer strings efficiently.
5. **Web Application**: Develop a web-based version using HTML, CSS, and Python.

The project can also be extended to compare different palindrome checking approaches, such as stack-and-queue processing, two-pointer comparison, and string reversal. Their execution time and memory usage can then be studied to understand algorithm efficiency.

9. CONCLUSION

The Palindrome Checker using Stack and Queue project is developed using Python to check whether a given word or number reads the same forward and backward. The system accepts input from the user, stores its characters in a Stack and a Queue, and compares the characters to determine whether the input is a palindrome.

This project demonstrates the practical use of two important data structures: Stack, which follows LIFO (Last In, First Out), and Queue, which follows FIFO (First In, First Out). By using these structures, students can understand how characters are inserted, removed, and compared during the checking process.

The program displays a clear result indicating whether the input is a palindrome or not. It also allows users to check multiple words or numbers in a single execution, making it simple and convenient to use.

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11. REFERENCE

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